

Singleton Energy and Factor-Exchange Collisions in Fixed-Shift Prime Fibers: Exact Undeleted Floors, Orbit Sparsity, and the Occupancy Barrier

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

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Abstract

The literal fixed-shift residual output is a coherent sum over distinct terminal primes inside each complete fiber $y = (\gamma, s, j)$. TPC-56 extracted a constant-energy phase sector, but that post-terminal adaptive selection did not lower-bound the undeleted output. We develop a phase-blind route that acts on no selected coordinates: the terminal diagonal carried by actual singleton fibers is already an exact positive part of the full grouped square.

For an arbitrary finite fiber system, let D be the diagonal energy, G the undeleted grouped energy, and D_1 the energy on fibers with exactly one active coordinate. We prove

$$G \geq D_1 \geq (D - I_1)_+,$$

where the rooted collision incidence is

$$I_1 = \sum_u |x_u|^2 (n_{q(u)} - 1).$$

More generally, rooted collision moments $I_r = \sum_u |x_u|^2 \binom{n_{q(u)}-1}{r}$ give an exact finite inclusion-exclusion formula for D_1 and sharp Bonferroni bounds. Thus the sufficient estimate $I_1 \leq (1 - \delta)D$ yields the undeleted lower square $G \geq \delta D$, without a transversal, a phase sector, or a monotonic return from a selected packet. A core cofactor block can be used directly: every collision partner lies in a fixed enlarged block.

For the literal bounded source class, write each terminal amplitude as $x_u = c_u \zeta_{m(u)}$ and form the baseline weighted row-fiber graph. We compute the sharp coefficient-uniform singleton constant

$$\beta_X = \min_{m: W_m > 0} \frac{W_{m,1}}{W_m} = \inf_{\substack{0 < D(\zeta) \\ |\zeta_m| \leq C}} \frac{D_1^{\text{phys}}(\zeta)}{D(\zeta)}.$$

This identifies rowwise singleton occupancy, rather than unweighted global abundance, as the exact uniform gate.

We then parameterize every literal collision. Two colliding factorizations have a unique factor exchange

$$d_1 = ab, \quad d_2 = ac, \quad r_1 = ce, \quad r_2 = be,$$

with a, b, c, e pairwise coprime and squarefree, and satisfy

$$\frac{e}{(e, h_0)} \mid b\ell_1 - c\ell_2.$$

Same-divisor collisions disappear for $r > \text{rad } |h_0|L$; high-cofactor cross-divisor collisions require an exchange of two comparable large factors. A fixed row pair and exchange pattern

occupies one orbit residue class. Separately, after (s, j, d) is fixed, the source and terminal primes run along two nonproportional affine forms.

These results reduce singleton occupancy to a concrete weighted one-atom versus two-atom incidence estimate with all source and terminal primes retained. The current carrier and coefficient hypotheses do not prove a positive value of β_X or a positive singleton proportion. Terminal activation, a parity improvement, and a twin-prime conclusion are not claimed.

Keywords: singleton fiber; rooted collision incidence; factor exchange; terminal prime; fixed physical shift; occupancy barrier.

Convention. One nonzero physical shift h_0 , one left source system, and one direct residue/orbit slice are fixed. Every domain and output energy uses the raw counting-measure atomic normalization inherited from TPC-53. The shaped terminal weight $\Lambda(n) = (\log n)\Phi_0(n/X)$ is displayed in the operator and is not the von Mangoldt function. A *singleton* is always defined relative to the actually nonzero full terminal support specified at the point of use.

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1 Introduction

The preceding four stages separated the fixed-shift prime square into auditable gates. TPC-53 fixed the literal coefficient dictionary and raw atomic normalization [3]. TPC-54 separated pre-terminal energy selection from actual terminal occupancy [4]. TPC-55 identified the complete residual fiber and proved that adjoining the terminal prime removes every collision before the final coherent sum [5]. TPC-56 then showed that a phase-adaptive pentant contains a constant fraction of the terminal diagonal and has a uniformly positive grouped square [6].

The last result is exact, but it deliberately deletes four phase channels. It therefore does not answer the undeleted question. If

$$x_{y,p} := (D_{h_0}G_X^L)_{u(y,p)} \quad (1)$$

is the terminal amplitude indexed by the exact terminal prime p in the full output fiber $y = (\gamma, s, j)$, then the two physical energies are

$$\mathcal{D}_{h_0}(G_X^L) = \sum_y \sum_{p \in \mathcal{P}_y^{\text{phys}}} |x_{y,p}|^2, \quad \mathcal{G}_{h_0}(G_X^L) = \sum_y \left| \sum_{p \in \mathcal{P}_y^{\text{phys}}} x_{y,p} \right|^2. \quad (2)$$

The first quantity ignores interference; the second is the actual undeleted grouped square. Arbitrary phases can make one nonsingleton fiber vanish. Neither bounded multiplicity nor a large collision-free selected rank prevents the omitted coordinates from cancelling it.

This paper opens a phase-blind door that does not delete any coordinate:

How much terminal diagonal energy already lies in fibers containing exactly one active atom, and what arithmetic event prevents an atom from being such a singleton?

On a singleton fiber there is no interference to estimate. Its grouped square equals its diagonal exactly, while every other grouped square is nonnegative. Thus singleton energy embeds monotonically in the full output even though a selected block or rank output does not.

1.1 The rooted collision interface

For a finite fiber map $q : \mathcal{U} \rightarrow \mathcal{Y}$, an array x , and active multiplicities

$$n_y(x) = \#\{u \in q^{-1}(y) : x_u \neq 0\}, \quad (3)$$

write D for the diagonal energy, G for the grouped energy, and D_1 for the energy on fibers with $n_y = 1$. The first rooted collision incidence is

$$I_1(x) = \sum_{u: x_u \neq 0} |x_u|^2 (n_{q(u)}(x) - 1) = \sum_{\substack{u \neq v \\ q(u)=q(v)}} |x_u|^2. \quad (4)$$

We prove the sharp chain

$$\boxed{G \geq D_1 \geq (D - I_1)_+}. \quad (5)$$

The first inequality uses the undeleted output itself. The second is the first Bonferroni truncation of an exact collision-moment expansion. Hence the concrete estimate

$$I_1(x) \leq (1 - \delta)D(x) \quad (6)$$

would imply $G(x) \geq \delta D(x)$. This is a genuine return to the full fixed- h_0 output, not a statement about a favorable subpacket.

The block version avoids another common logical error. Let the root atom lie in a cofactor core $R \leq r \leq 2R$, but count all of its collision partners in the full active support. Because the direct squarefree factor always lies in $[D, 2D]$, equality $d_u r_u = d_v r_v$ forces every partner into $R/2 \leq r_v \leq 4R$. Thus a core-block singleton estimate can be proved from one fixed enlarged block and then enters the full output directly. No lower monotonicity of grouped norms is invoked.

1.2 Why unweighted singleton counts are insufficient

The inherited source hypothesis has the form

$$x_u = c_u \zeta_{m(u)}, \quad |\zeta_m| \leq C, \quad (7)$$

where c_u contains every literal mask, profile, terminal, and common fiber factor independent of ζ_m . One source row may feed many different fibers. Therefore a large unweighted set of singleton atoms does not give a coefficient-uniform energy proportion: the allowed source energy can concentrate on rows meeting mostly nonsingleton fibers.

We encode the exact obstruction in a weighted bipartite graph. Its left vertices are source rows, its right vertices are baseline terminal fibers, and its edges are baseline-nondegenerate terminal atoms with weights $|c_u|^2$. If W_m is the total edge weight from row m and $W_{m,1}$ is the part ending in structurally singleton fibers, define

$$\beta_X := \min_{m: W_m > 0} \frac{W_{m,1}}{W_m}. \quad (8)$$

The exact minimax theorem is

$$\boxed{\inf_{\substack{0 < D(\zeta) \\ |\zeta_m| \leq C}} \frac{D_1^{\text{phys}}(\zeta)}{D(\zeta)} = \beta_X}. \quad (9)$$

Thus a uniform Door-A theorem is equivalent to a rowwise weighted occupancy statement. If $\beta_X = 0$, singleton energy cannot by itself give a coefficient-uniform positive fraction. This does not say that the full grouped energy vanishes, nor does it say that the distinguished source coefficients concentrate on a minimizing row.

1.3 Every collision is a factor exchange

The arithmetic side is equally explicit. Suppose two distinct terminal atoms in one fiber have

$$m_i = d_i \ell_i, \quad d_1 r_1 = d_2 r_2 = s. \quad (10)$$

The inherited equality support makes s squarefree and $(d_i, r_i) = 1$. Consequently there are unique pairwise coprime squarefree integers a, b, c, e such that

$$d_1 = ab, \quad d_2 = ac, \quad r_1 = ce, \quad r_2 = be. \quad (11)$$

The two terminal equations become

$$ce p_1 - ab \ell_1 j = h_0, \quad be p_2 - ac \ell_2 j = h_0, \quad (12)$$

and force

$$\boxed{\frac{e}{(e, h_0)} \mid b \ell_1 - c \ell_2.} \quad (13)$$

This is not an ambient divisor model: the reverse statement retains every literal carrier, primality, mask, cutoff, residue, and orbit condition.

The factor exchange gives deterministic sparsity. Same- d collisions are impossible for $r > \text{rad } |h_0| L$. At large cofactor scale, a cross- d collision must exchange two distinct comparable factors of quantified size. After the rows and exchange pattern are fixed, simultaneous integrality occupies one residue class of j . In the separate fixed- (s, j, d) description, the source and terminal primes run along two nonproportional affine forms. These are the exact orbit and two-point objects that a future upper-bound sieve would have to sum.

1.4 What advances and what remains open

The advance is a complete reduction of the phase-blind undeleted route:

$$\begin{aligned} &\text{terminal activation} + \text{weighted rooted-collision deficit} \\ &\implies \text{undeleted fixed-}h_0 \text{ lower square.} \end{aligned}$$

All collisions in the numerator now have a unique factor-exchange and orbit-ladder description. The remaining estimate compares weighted one-atom terminal energy with a weighted two-atom incidence sum, retaining the two source primes and the two terminal primes in every collision.

The current TPC-53-56 hypotheses do not imply (6). The retained carrier has no declared product-density theorem, terminal activation is itself open, a maximum-degree bound does not force any singleton, and the bounded source class permits rowwise energy concentration. An upper-bound sieve for the two affine prime forms can estimate a numerator, but a positive proportion requires a compatible lower denominator and uniform control of the carrier and weights. This is precisely where the next arithmetic input is needed.

[Section 2](#) fixes the literal support and energies. [Section 3](#) proves the finite collision-moment theory. [Section 4](#) gives the full-output block bridge. [Section 5](#) proves the rowwise minimax theorem. [Section 6](#) parameterizes every collision, and [section 7](#) resolves its orbit ladder. [Section 8](#) states the exact missing estimate and the claim boundary.

The finite-fiber identities are unconditional mathematics (L0). Their specialization to the exact terminal-prime carrier and the factor-exchange structure are literal fixed-shift interface results (L1). No positive singleton proportion, terminal-activation theorem, parity-sensitive sieve gain, or twin-prime conclusion is asserted; the classical parity boundary remains in force [1].

2 The literal fixed-shift terminal-prime interface

The singleton argument is meaningful only for the exact physical synthesis. This section therefore fixes the input atoms, terminal multiplier, output key, and counting normalization before any collision statistic is formed. The definitions are inherited from TPC-53, TPC-55, and TPC-56 [3, 5, 6]; they are repeated here so that an ambient prime ladder, a renormalized model, or a selected output cannot be silently substituted for the literal map.

2.1 Rows, masks, and pre-terminal atoms

Fix one integer shift

$$h_0 \in \mathbb{Z} \setminus \{0\}, \quad H := \text{rad } |h_0|, \quad (14)$$

one left source system, one direct residue group, and one inherited residue/orbit slice. A direct row and an opposite row have the unique displayed factorizations

$$m = d_m \ell_m, \quad m_\gamma = d_\gamma \ell_\gamma, \quad (15)$$

where

$$\ell_m, \ell_\gamma \in [L, 2L] \text{ are prime,} \quad d_m, d_\gamma \in [D, 2D] \text{ are squarefree,} \quad (d_m d_\gamma, H) = 1. \quad (16)$$

The inherited scale relations are

$$Q = LD, \quad K = DJ, \quad QJ \asymp X, \quad \Delta = QX^{-1/400}, \quad G = X^{1/400}. \quad (17)$$

All fixed compact boxes, residue classes, and smooth-function families below are the declared ones from the parent packet; none is enlarged after the terminal equation is imposed.

The exact direct source coefficient is

$$a_m = \mu(d_m)(\log \ell_m) \omega_D(d_m) \psi_L(\ell_m/L) \zeta_m, \quad |\zeta_m| \ll 1. \quad (18)$$

No lower bound, nonvanishing assumption, or phase condition is imposed on ζ_m . The opposite row contributes

$$\eta_\gamma \mathcal{R}(q_\gamma), \quad q_\gamma = \left(\log \frac{\ell_\gamma}{L}, \log \frac{d_\gamma}{D} \right), \quad |\eta_\gamma| = 1, \quad (19)$$

and the fixed orbit class contributes $\xi_a \neq 0$. Retain the literal mask and four-factor profile

$$K_{m\gamma} := \mathbf{1}_{\ell_m \neq \ell_\gamma} \mathbf{1}_{|m-m_\gamma| > \Delta} \mathbf{1}_{(d_m, d_\gamma) \leq G}, \quad (20)$$

$$\mathcal{W}_{m,\gamma}(j/J) := W_1(mj/X) \Psi_1(d_m j/K) W_2(m_\gamma j/X) \Psi_2(d_\gamma j/K). \quad (21)$$

Let $\mathcal{I}_X$ be the exact retained set of (m, r) -pairs, $\mathcal{O}_X$ the complete opposite-row cell, and

$$\mathcal{J}_{X,a} := \{j : j/J \in I_0, j \equiv a \pmod{H_{\text{orb}}}\}, \quad H_{\text{orb}} = O_{h_0}(1). \quad (22)$$

Here $I_0 \Subset (0, \infty)$ is fixed, so every retained time satisfies $j \asymp J$. The full pre-terminal universe and the distinguished lifted vector are

$$\mathcal{U}^{\text{pre}} := \{(m, r, j, \gamma) : (m, r) \in \mathcal{I}_X, j \in \mathcal{J}_{X,a}, \gamma \in \mathcal{O}_X\}, \quad (23)$$

$$[G_X^L]_u := a_m K_{m\gamma} \eta_\gamma \mathcal{R}(q_\gamma) \xi_a \mathcal{W}_{m,\gamma}(j/J), \quad u = (m, r, j, \gamma). \quad (24)$$

The scalar in (24) has no explicit r -dependence. Its repetitions over distinct retained cofactors are nevertheless distinct counting-measure atoms.

2.2 Terminal support and the complete output key

Let

$$\Lambda(n) := (\log n)\Phi_0(n/X). \quad (25)$$

This is a shaped terminal logarithm, not the von Mangoldt function. For $u = (m, r, j, \gamma) \in \mathcal{U}^{\text{pre}}$, define

$$p_u := \frac{mj + h_0}{r} \quad (26)$$

when the quotient is integral. The terminal-admissible universe is exactly

$$\begin{aligned} \mathcal{U}_{h_0}^{\text{term}} := \{u \in \mathcal{U}^{\text{pre}} : p_u \text{ is prime in the inherited terminal support,} \\ p_u r - mj = h_0, \quad \Lambda(p_u r) \neq 0, \\ d_m, r, d_m r \text{ are squarefree, } (d_m, r) = 1\}. \end{aligned} \quad (27)$$

The actually nonzero pre-terminal support is kept separate:

$$\mathcal{U}_{h_0}^{\text{phys}} := \{u \in \mathcal{U}_{h_0}^{\text{term}} : [G_X^L]_u \neq 0\}. \quad (28)$$

On $\mathcal{U}_{h_0}^{\text{term}}$, writing $m = d_m \ell_m$ and $s = d_m r$, one has

$$p_u r - d_m \ell_m j = h_0, \quad s \text{ squarefree, } (d_m, r) = 1, \quad \mu(d_m)\mu(r) = \mu(s). \quad (29)$$

The complete physical key and its prime-resolved refinement are

$$q(u) := (\gamma, d_m r, j), \quad \tilde{q}(u) := (\gamma, d_m r, j, p_u). \quad (30)$$

For $y \in q(\mathcal{U}_{h_0}^{\text{term}})$, define

$$\mathcal{F}_y^{\text{term}} := q^{-1}(y) \cap \mathcal{U}_{h_0}^{\text{term}}, \quad \mathcal{F}_y^{\text{phys}} := q^{-1}(y) \cap \mathcal{U}_{h_0}^{\text{phys}}. \quad (31)$$

Proposition 2.1 (Inherited prime-resolved injectivity). *Assume $D = o(J)$, uniqueness of the row factorization (15), and sufficiently large X . On the one declared source system, $\tilde{q}$ is injective on $\mathcal{U}_{h_0}^{\text{term}}$. Consequently, within a fixed complete output fiber, distinct terminal atoms have distinct source rows and distinct terminal primes.*

Proof. Suppose $\tilde{q}(u_1) = \tilde{q}(u_2)$, and write $m_i = d_i \ell_i$. The two atoms have the same s, j, p, γ , with $r_i = s/d_i$. Multiplying $pr_i = d_i \ell_i j + h_0$ by d_i gives

$$ps = d_i^2 \ell_i j + h_0 d_i.$$

Hence

$$j(\ell_1 d_1^2 - \ell_2 d_2^2) = h_0(d_2 - d_1). \quad (32)$$

If the integer in parentheses is nonzero, the left side has size $\gg J$, whereas the right side is $O_{h_0}(D) = o(J)$. Thus both sides vanish. Since $h_0 \neq 0$, one gets $d_1 = d_2$, then $\ell_1 = \ell_2$, and finally $r_1 = r_2$. The two atoms coincide. The assertions about rows and primes follow. $\square$

2.3 Terminalization, grouping, and raw energies

For an arbitrary array $g = (g_u)_{u \in \mathcal{U}^{\text{pre}}}$, define the terminal diagonal by

$$(D_{h_0} g)_u := \begin{cases} \mu(r_u) \Lambda(p_u r_u) g_u, & u \in \mathcal{U}_{h_0}^{\text{term}}, \\ 0, & u \in \mathcal{U}^{\text{pre}} \setminus \mathcal{U}_{h_0}^{\text{term}}. \end{cases} \quad (33)$$

The literal grouped map is

$$(\mathcal{T}_{h_0} g)_y := \sum_{u \in \mathcal{F}_y^{\text{term}}} (D_{h_0} g)_u. \quad (34)$$

Every atomic Hilbert space uses raw counting measure. Thus

$$\mathcal{E}_{\text{pre}}(g) := \sum_{u \in \mathcal{U}^{\text{pre}}} |g_u|^2, \quad (35)$$

$$\mathcal{D}_{h_0}(g) := \|D_{h_0}g\|_2^2 = \sum_{u \in \mathcal{U}_{h_0}^{\text{term}}} |\Lambda(p_u r_u)|^2 |g_u|^2, \quad (36)$$

$$\mathcal{G}_{h_0}(g) := \|\mathcal{T}_{h_0}g\|_2^2 = \sum_y \left| \sum_{u \in \mathcal{F}_y^{\text{term}}} (D_{h_0}g)_u \right|^2. \quad (37)$$

The second identity uses squarefreeness of r_u on the terminal support. The terminal multiplier remains inside D_{h_0} ; no factor such as $1/(E_X J)$ is inserted into any of these norms.

For the distinguished vector, put

$$x_u := (D_{h_0} G_X^L)_u. \quad (38)$$

If $u = (d\ell, r, j, \gamma) \in \mathcal{U}_{h_0}^{\text{term}}$ and $y = q(u) = (\gamma, s, j)$, define

$$\Omega_y := \mu(s) \eta_\gamma \mathcal{R}(q_\gamma) \xi_a W_2(m_\gamma j/X) \Psi_2(d_\gamma j/K), \quad (39)$$

$$\kappa_u := (\log \ell) \omega_D(d) \psi_L(\ell/L) K_{m_\gamma} W_1(mj/X) \Psi_1(dj/K) \Lambda(mj + h_0). \quad (40)$$

The Möbius identity in (29) gives the exact factorization

$$\boxed{x_u = \Omega_{q(u)} \kappa_u \zeta_m = c_u \zeta_m, \quad c_u := \Omega_{q(u)} \kappa_u.} \quad (41)$$

Thus c_u contains every displayed literal factor except the bounded distinguished row coefficient ζ_m . Since the terminal multiplier is nonzero on $\mathcal{U}_{h_0}^{\text{term}}$,

$$x_u \neq 0 \iff [G_X^L]_u \neq 0 \iff u \in \mathcal{U}_{h_0}^{\text{phys}}. \quad (42)$$

2.4 Hostile interface audit

The following invariants are part of every subsequent statement.

- (I1) *One literal coefficient.* Collision energies use $|x_u|^2$ from (38), not an unweighted count and not a coefficient envelope.
- (I2) *One fixed shift.* The equality is always $p_u r_u - m_u j = h_0$ for the same specified $h_0 \neq 0$. No average over shifts, frozen intercept, or shift-dependent reselection is part of the interface.
- (I3) *Three different supports.* The containments

$$\mathcal{U}_{h_0}^{\text{phys}} \subseteq \mathcal{U}_{h_0}^{\text{term}} \subseteq \mathcal{U}^{\text{pre}}$$

are not identities. In particular, ambient solutions of the determinant equation are not automatically terminal or physical atoms.

- (I4) *Complete fibers.* Singleton multiplicity uses the full key $q(u) = (\gamma, d_m r, j)$ and every active terminal atom. Restricting a cofactor block, phase sector, rank, or prime interval changes the fiber system and cannot be undone by monotonicity of a coherent square.
- (I5) *Prime resolution.* The refined key $\tilde{q}$ is injective, but the physical output forgets p_u and coherently adds all primes in one q -fiber. The prime-resolved diagonal therefore does not itself lower-bound the undeleted grouped output.

- (I6) *Raw ledger.* Repeated cofactor labels remain separate counting-measure atoms. Selection, cofactor-block occupancy, terminal activation, singleton occupancy, native-Gram comparison, boundary control, and tail reconnection must each be charged separately before any endpoint conclusion is stated.

Accordingly, the literal interface supplies exact identities and support geometry only. It does not assert that a chosen block is nonempty, that terminal diagonal energy is comparable to pre-terminal energy, that a positive fraction of physical fibers are singletons, or that the undeleted grouped square has an asymptotic lower bound.

3 Singleton energy and collision moments

This section isolates the finite-dimensional statement needed for the undeleted terminal output. No arithmetic, phase hypothesis, or choice of a subpacket is used. Let $\mathcal{U}$ and $\mathcal{Y}$ be finite sets, let $q : \mathcal{U} \rightarrow \mathcal{Y}$ be a fiber map, and write

$$\mathcal{F}_y := q^{-1}(y). \quad (43)$$

For $x = (x_u)_{u \in \mathcal{U}} \in \mathbb{C}^{\mathcal{U}}$, its active support and its active fiber multiplicities are

$$\mathcal{U}(x) := \{u \in \mathcal{U} : x_u \neq 0\}, \quad n_y(x) := |\mathcal{F}_y \cap \mathcal{U}(x)|. \quad (44)$$

The diagonal and undeleted grouped energies are

$$D(x) := \sum_{u \in \mathcal{U}} |x_u|^2, \quad G(x) := \sum_{y \in \mathcal{Y}} \left| \sum_{u \in \mathcal{F}_y} x_u \right|^2. \quad (45)$$

Define the physical singleton energy by

$$D_1(x) := \sum_{\substack{y \in \mathcal{Y} \\ n_y(x)=1}} \sum_{u \in \mathcal{F}_y} |x_u|^2. \quad (46)$$

The word “physical” here refers to the active support of the one vector x : a fiber that becomes a singleton because some of its ambient coordinates vanish is counted in (46).

Theorem 3.1 (Undeleted singleton floor). *For every finite fiber system and every $x \in \mathbb{C}^{\mathcal{U}}$,*

$$\boxed{G(x) \geq D_1(x)}. \quad (47)$$

This is a lower bound for the complete grouped vector; no coordinate is deleted before grouping.

Proof. If $n_y(x) = 1$, there is a unique $u_y \in \mathcal{F}_y$ with $x_{u_y} \neq 0$, and hence

$$\left| \sum_{u \in \mathcal{F}_y} x_u \right|^2 = |x_{u_y}|^2.$$

Every remaining summand in $G(x)$ is nonnegative. Summing the displayed identity over the singleton fibers proves (47). $\square$

We next encode singleton occupancy by rooted collision moments. Adopt the convention $\binom{a}{b} = 0$ when $b > a$, and define

$$I_r(x) := \sum_{u \in \mathcal{U}(x)} |x_u|^2 \binom{n_{q(u)}(x) - 1}{r}, \quad r \geq 0. \quad (48)$$

Thus $I_0(x) = D(x)$, while

$$I_1(x) = \sum_{u \in \mathcal{U}(x)} |x_u|^2 (n_{q(u)}(x) - 1) = \sum_{\substack{u, v \in \mathcal{U}(x) \\ u \neq v, q(u)=q(v)}} |x_u|^2. \quad (49)$$

The last sum is over ordered collision pairs, with the energy placed on the root u .

Theorem 3.2 (Exact collision-moment expansion). *Suppose $x \neq 0$, and put*

$$M(x) := \max_{y \in \mathcal{Y}} n_y(x). \quad (50)$$

Then

$$D_1(x) = \sum_{r=0}^{M(x)-1} (-1)^r I_r(x). \quad (51)$$

More generally, every admissible odd truncation is a lower bound and every admissible even truncation is an upper bound:

$$\sum_{r=0}^{2k+1} (-1)^r I_r(x) \leq D_1(x) \quad \text{if } 2k+1 \leq M(x)-1, \quad (52)$$

$$D_1(x) \leq \sum_{r=0}^{2k} (-1)^r I_r(x) \quad \text{if } 2k \leq M(x)-1. \quad (53)$$

In particular,

$$G(x) \geq D_1(x) \geq D(x) - I_1(x), \quad G(x) \geq (D(x) - I_1(x))_+. \quad (54)$$

Proof. For an active coordinate u , set $N = n_{q(u)}(x) - 1$. The binomial identity

$$\sum_{r=0}^N (-1)^r \binom{N}{r} = (1-1)^N = \mathbf{1}_{\{N=0\}} \quad (55)$$

shows, after multiplication by $|x_u|^2$ and summation over u , that the right-hand side of (51) equals $D_1(x)$.

For $N \geq 1$ and $0 \leq s < N$, the partial-sum identity

$$\sum_{r=0}^s (-1)^r \binom{N}{r} = (-1)^s \binom{N-1}{s} \quad (56)$$

has the sign asserted in (52)–(53); for $s \geq N$, the partial sum is already the exact value in (55). Summing again with the nonnegative weights $|x_u|^2$ proves both Bonferroni inequalities. If $M(x) = 1$, then $I_1(x) = 0$ and $D_1(x) = D(x)$ directly. Otherwise the case $s = 1$, together with theorem 3.1 and $G(x) \geq 0$, proves (54). $\square$

Proposition 3.3 (Sharpness of the first-moment floor). *The nonnegative lower bound in (54) is sharp using only the two numbers D and I_1 . More precisely, for every $D_0 > 0$ and $I_0 \geq 0$ there is a finite fiber system and a vector x such that*

$$D(x) = D_0, \quad I_1(x) = I_0, \quad G(x) = (D_0 - I_0)_+. \quad (57)$$

Proof. Suppose first that $I_0 \leq D_0$. Use one singleton fiber carrying a coordinate of squared modulus $D_0 - I_0$, omitting it when $I_0 = D_0$. If $I_0 > 0$, use one further fiber with two coordinates

$$x_{u_1} = \sqrt{I_0/2}, \quad x_{u_2} = -\sqrt{I_0/2}.$$

The two-atom fiber has diagonal energy and first rooted collision moment equal to I_0 , and its grouped energy is zero. This proves the claim in the stated range.

Now suppose $I_0 > D_0$, write

$$\frac{I_0}{D_0} = k + \theta, \quad k \in \mathbb{N}, \quad 0 \leq \theta < 1,$$

and use two fibers of degrees $k + 1$ and $k + 2$, with total diagonal energies $(1 - \theta)D_0$ and θD_0 , respectively. Omit the second fiber when $\theta = 0$. Within each nonzero fiber, take equal coordinate magnitudes and phases proportional to all roots of unity of the corresponding degree. Each fiber sum is zero. A fiber of degree n contributes $(n - 1)$ times its diagonal energy to I_1 , and therefore the two fibers give

$$I_1(x) = (k(1 - \theta) + (k + 1)\theta)D_0 = (k + \theta)D_0 = I_0.$$

They have total diagonal energy D_0 and grouped energy zero, completing the proof. $\square$

The following localized form is the deterministic bridge used later for a cofactor block. It is formulated for an arbitrary coordinate subset so that no monotonicity of grouped norms is being assumed.

Proposition 3.4 (Core-to-full collision bridge). *Let $\mathcal{B} \subseteq \mathcal{U}$, and define*

$$D_{\mathcal{B}}(x) := \sum_{u \in \mathcal{B}} |x_u|^2, \tag{58}$$

$$I_{\mathcal{B}}^{\text{full}}(x) := \sum_{u \in \mathcal{B} \cap \mathcal{U}(x)} |x_u|^2 (n_{q(u)}(x) - 1). \tag{59}$$

Here the multiplicity in (59) is computed in the full active support $\mathcal{U}(x)$, not merely in $\mathcal{B}$. Then

$$\boxed{G(x) \geq (D_{\mathcal{B}}(x) - I_{\mathcal{B}}^{\text{full}}(x))_+}. \tag{60}$$

Consequently, if $D_{\mathcal{B}}(x) > 0$ and, for some $0 < \delta \leq 1$,

$$I_{\mathcal{B}}^{\text{full}}(x) \leq (1 - \delta)D_{\mathcal{B}}(x), \tag{61}$$

then the complete undeleted output satisfies

$$G(x) \geq \delta D_{\mathcal{B}}(x). \tag{62}$$

Proof. Let

$$E_{\mathcal{B}}^{\text{mult}}(x) := \sum_{\substack{u \in \mathcal{B} \cap \mathcal{U}(x) \\ n_{q(u)}(x) \geq 2}} |x_u|^2.$$

Every active coordinate in $\mathcal{B}$ not counted by $E_{\mathcal{B}}^{\text{mult}}(x)$ belongs to a singleton of the full fiber system. Hence [theorem 3.1](#) gives

$$G(x) \geq D_{\mathcal{B}}(x) - E_{\mathcal{B}}^{\text{mult}}(x).$$

For every term in $E_{\mathcal{B}}^{\text{mult}}(x)$, the integer $n_{q(u)}(x) - 1$ is at least one. Therefore

$$E_{\mathcal{B}}^{\text{mult}}(x) \leq I_{\mathcal{B}}^{\text{full}}(x),$$

which proves (60). Substitution of (61) proves (62). $\square$

Remark 3.5 (Exact scope). The preceding results do not assert that singleton energy is present in a particular arithmetic packet. They identify a sufficient statistic for the undeleted output: a positive relative upper bound for the rooted collision incidence. Estimating that statistic is an additional arithmetic task.

4 A cofactor-core bridge to the complete physical output

The singleton floor becomes useful only after two support issues are kept separate. First, a cofactor block is a set of input coordinates, whereas the target norm is the complete undeleted output. A lower bound for the grouped norm of the restricted block is not monotone under restoring the omitted coordinates. Second, whether a terminal-admissible coordinate is physically active depends on the distinguished source coefficient. This section inserts both distinctions into the literal fixed-shift map and identifies the exact weighted collision estimate that would close the singleton route.

4.1 The literal terminalized vector and a cofactor core

Fix $h_0 \neq 0$, one declared left source system, and the direct residue/orbit slice inherited from TPC-53-56 [3, 5, 6]. We use the exact terminal and physical sets

$$\mathcal{U}_{h_0}^{\text{phys}} \subseteq \mathcal{U}_{h_0}^{\text{term}} \subseteq \mathcal{U}^{\text{pre}}$$

and the complete output key

$$q(u) = (\gamma, d_m r, j), \quad u = (m, r, j, \gamma), \quad (63)$$

from TPC-55. Put

$$x_u := (D_{h_0} G_X^L)_u, \quad u \in \mathcal{U}_{h_0}^{\text{term}}. \quad (64)$$

By definition of $\mathcal{U}_{h_0}^{\text{phys}}$, and because the terminal multiplier $\mu(r_u)\Lambda(p_u r_u)$ is nonzero on $\mathcal{U}_{h_0}^{\text{term}}$, one has

$$x_u \neq 0 \iff [G_X^L]_u \neq 0 \iff u \in \mathcal{U}_{h_0}^{\text{phys}}. \quad (65)$$

Thus the active multiplicity of a physical output fiber is

$$n_y^{\text{phys}} := \left| q^{-1}(y) \cap \mathcal{U}_{h_0}^{\text{phys}} \right|. \quad (66)$$

All norms below use raw counting measure. In particular,

$$\mathcal{D}_{h_0}(G_X^L) = \sum_{u \in \mathcal{U}_{h_0}^{\text{term}}} |x_u|^2, \quad (67)$$

$$\mathcal{G}_{h_0}(G_X^L) = \sum_y \left| \sum_{u \in q^{-1}(y)} x_u \right|^2. \quad (68)$$

The sum in (68) is the original undeleted fixed- h_0 synthesis, not a selected packet.

For a cofactor scale $R > 0$, define the pre-terminal and terminal core sets

$$\mathcal{B}_R^{\text{pre}} := \{u \in \mathcal{U}^{\text{pre}} : R \leq r_u \leq 2R\}, \quad (69)$$

$$\mathcal{B}_R^{\text{term}} := \mathcal{B}_R^{\text{pre}} \cap \mathcal{U}_{h_0}^{\text{term}}. \quad (70)$$

Its terminal diagonal energy is

$$\mathcal{D}_R^{\text{core}} := \sum_{u \in \mathcal{B}_R^{\text{term}}} |x_u|^2. \quad (71)$$

The rooted collision incidence of the core against the *full* physical vector is

$$\mathcal{I}_R^{\text{full}} := \sum_{u \in \mathcal{B}_R^{\text{term}} \cap \mathcal{U}_{h_0}^{\text{phys}}} |x_u|^2 (n_{q(u)}^{\text{phys}} - 1). \quad (72)$$

The multiplicity in (72) is not computed in the restricted block. This is the essential point that makes the resulting lower bound a statement about the complete output.

Theorem 4.1 (Full-output cofactor-core singleton bridge). *For the distinguished literal terminal vector,*

$$\boxed{\mathcal{G}_{h_0}(G_X^L) \geq (\mathcal{D}_R^{\text{core}} - \mathcal{I}_R^{\text{full}})_+}. \quad (73)$$

Consequently, if $\mathcal{D}_R^{\text{core}} > 0$ and

$$\mathcal{I}_R^{\text{full}} \leq (1 - \delta_R)\mathcal{D}_R^{\text{core}} \quad (0 < \delta_R \leq 1), \quad (74)$$

then the complete undeleted output satisfies

$$\boxed{\mathcal{G}_{h_0}(G_X^L) \geq \delta_R \mathcal{D}_R^{\text{core}}}. \quad (75)$$

Proof. Apply [proposition 3.4](#) to the full terminalized vector (64) and the coordinate subset $\mathcal{B}_R^{\text{term}}$. Equations (65) and (66) identify the abstract full active multiplicity with n_y^{phys} , giving (73). Substitution of (74) gives (75). $\square$

No inequality between $\mathcal{G}_{h_0}(G_X^L \mathbf{1}_{\mathcal{B}_R^{\text{pre}}})$ and $\mathcal{G}_{h_0}(G_X^L)$ is used in the proof. Coordinates outside the core may cancel a restricted block sum. The theorem avoids that invalid monotonicity step by retaining only core coordinates that are singleton in the full active fiber system.

4.2 Every partner lies in one enlarged cofactor block

Although (72) uses full active multiplicities, its possible partners are geometrically local. Define

$$\mathcal{B}_R^+ := \{u \in \mathcal{U}_{h_0}^{\text{term}} : R/2 \leq r_u \leq 4R\}, \quad (76)$$

$$n_y^+(R) := \left| q^{-1}(y) \cap \mathcal{U}_{h_0}^{\text{phys}} \cap \mathcal{B}_R^+ \right|. \quad (77)$$

Lemma 4.2 (Factor-exchange localization). *Let $u \in \mathcal{B}_R^{\text{term}}$, and suppose $v \in \mathcal{U}_{h_0}^{\text{term}}$ satisfies $q(v) = q(u)$. Then*

$$R/2 \leq r_v \leq 4R. \quad (78)$$

In particular, if a physical fiber meets the core, then its complete physical multiplicity is already visible in the enlarged block:

$$n_{q(u)}^{\text{phys}} = n_{q(u)}^+(R) \quad (u \in \mathcal{B}_R^{\text{term}} \cap \mathcal{U}_{h_0}^{\text{phys}}). \quad (79)$$

Proof. Write

$$u = (d\ell, r, j, \gamma), \quad v = (d'\ell', r', j, \gamma).$$

Equality of the complete output keys gives the exact factor exchange

$$dr = d'r'. \quad (80)$$

Both source squarefree factors lie in the inherited dyadic interval $[D, 2D]$. Hence

$$r' = \frac{d}{d'}r, \quad \frac{1}{2} \leq \frac{d}{d'} \leq 2.$$

Together with $R \leq r \leq 2R$, this proves (78). Every full physical partner of a physical core atom is therefore counted by (77), proving (79). $\square$

The constants $1/2$ and 4 come only from the declared dyadic squarefree support. Thus enlarging the core creates an $O(1)$ cover and has no positive polynomial cost. More explicitly,

$$\boxed{\mathcal{I}_R^{\text{full}} = \sum_{\substack{u \in \mathcal{U}_{h_0}^{\text{phys}} \cap \mathcal{B}_R^{\text{term}} \\ v \in \mathcal{U}_{h_0}^{\text{phys}} \cap \mathcal{B}_R^+ \\ v \neq u, q(v)=q(u)}} |x_u|^2}. \quad (81)$$

This is an ordered pair sum with the terminal energy placed on the core root u . It is not an unweighted count of factor exchanges.

4.3 Row-factorized form of the exact missing estimate

The exact literal interface (41) factors every terminal amplitude as

$$x_u = \Omega_{q(u)} \kappa_u \zeta_{m_u}. \quad (82)$$

Here κ_u is the product of the shaped terminal logarithm, direct source shapes, mask, and every other displayed factor in the remaining amplitude except ζ_{m_u} . Put

$$c_u := \Omega_{q(u)} \kappa_u \quad (83)$$

and define the baseline-nondegenerate carrier

$$\mathcal{U}_{h_0}^\circ := \{u \in \mathcal{U}_{h_0}^{\text{term}} : c_u \neq 0\}. \quad (84)$$

Then

$$x_u = c_u \zeta_{m_u}, \quad u \in \mathcal{U}_{h_0}^\circ, \quad (85)$$

and a baseline edge is physically active precisely when its row coefficient is nonzero. Distinct terminal atoms in a fixed output fiber have distinct source rows by the prime-resolved injection of TPC-55. Thus a fixed pair (m, y) supports at most one baseline edge, as required by [theorem 5.1](#).

For each row define its core baseline energy

$$W_{m,R} := \sum_{\substack{u \in \mathcal{U}_{h_0}^\circ \cap \mathcal{B}_R^{\text{term}} \\ m_u = m}} |c_u|^2. \quad (86)$$

On the enlarged baseline carrier put

$$d_y^+(R) := \left| q^{-1}(y) \cap \mathcal{U}_{h_0}^\circ \cap \mathcal{B}_R^+ \right|, \quad (87)$$

and define the support-only rooted row incidence

$$J_{m,1}^{R,+} := \sum_{\substack{u \in \mathcal{U}_{h_0}^\circ \cap \mathcal{B}_R^{\text{term}} \\ m_u = m}} |c_u|^2 (d_{q(u)}^+(R) - 1). \quad (88)$$

For the distinguished coefficient vector, let

$$n_y^+(R; \zeta) := \left| \{v \in q^{-1}(y) \cap \mathcal{U}_{h_0}^\circ \cap \mathcal{B}_R^+ : \zeta_{m_v} \neq 0\} \right| \quad (89)$$

and

$$\mathfrak{J}_{m,1}^{R,+}(\zeta) := \mathbf{1}_{\{\zeta_m \neq 0\}} \sum_{\substack{u \in \mathcal{U}_{h_0}^\circ \cap \mathcal{B}_R^{\text{term}} \\ m_u = m}} |c_u|^2 (n_{q(u)}^+(R; \zeta) - 1). \quad (90)$$

The row indicator makes the active-root restriction literal. In particular, an inactive row contributes zero rather than a spurious negative value from $n_{q(u)}^+(R; \zeta) - 1$.

Proposition 4.3 (Exact distinguished incidence and its baseline majorant). *The core diagonal and full rooted collision incidence satisfy*

$$\mathcal{D}_R^{\text{core}} = \sum_m |\zeta_m|^2 W_{m,R}, \quad (91)$$

$$\mathcal{I}_R^{\text{full}} = \sum_m |\zeta_m|^2 \mathfrak{J}_{m,1}^{R,+}(\zeta), \quad (92)$$

$$0 \leq \mathfrak{J}_{m,1}^{R,+}(\zeta) \leq J_{m,1}^{R,+}. \quad (93)$$

Consequently the coefficient-weighted support estimate

$$\sum_m |\zeta_m|^2 J_{m,1}^{R,+} \leq (1 - \delta_R) \sum_m |\zeta_m|^2 W_{m,R} \quad (94)$$

is sufficient for the undeleted lower bound (75).

Proof. Equation (91) follows by substituting (85) into (71). For an active core root u , lemma 4.2 identifies its full active degree with $n_{q(u)}^+(R; \zeta)$. Substitution into (72) and grouping the roots by their source row proves (92). Since active enlarged edges form a subset of the baseline enlarged carrier, $n_y^+(R; \zeta) \leq d_y^+(R)$, proving (93). Therefore (94) implies (74), and theorem 4.1 finishes the proof. $\square$

The exact weakest first-moment target is (92) with $\mathfrak{J}_{m,1}^{R,+}(\zeta)$. Replacing it by the baseline quantity $J_{m,1}^{R,+}$ is a stronger coefficient-independent sufficient condition. Neither target follows from a maximal-degree upper bound: a carrier may consist entirely of doubleton fibers.

4.4 Activation and the prospective loss ledger

To state the route without hiding an occupancy assertion, let $\mathcal{E}_{\text{par}}$ be the parent physical atomic energy before selection of the declared literal packet. Define the selected pre-terminal energy and the core pre-terminal energy by

$$\mathcal{E}_{\text{pre}}(G_X^L) := \sum_{u \in \mathcal{U}^{\text{pre}}} |[G_X^L]_u|^2, \quad (95)$$

$$\mathcal{E}_R^{\text{pre}}(G_X^L) := \sum_{u \in \mathcal{B}_R^{\text{pre}}} |[G_X^L]_u|^2. \quad (96)$$

Suppose, along a sequence of scales,

$$\mathcal{E}_{\text{pre}}(G_X^L) \geq X^{-\lambda_{\text{sel}} + o(1)} \mathcal{E}_{\text{par}}, \quad (97)$$

$$\mathcal{E}_R^{\text{pre}}(G_X^L) \geq X^{-\lambda_{\text{blk}} + o(1)} \mathcal{E}_{\text{pre}}(G_X^L), \quad (98)$$

$$\mathcal{D}_R^{\text{core}} \geq X^{-\lambda_{\text{term}} + o(1)} \mathcal{E}_R^{\text{pre}}(G_X^L), \quad (99)$$

$$\delta_R \geq X^{-\lambda_{\text{sing}} + o(1)}, \quad \mathcal{I}_R^{\text{full}} \leq (1 - \delta_R) \mathcal{D}_R^{\text{core}}. \quad (100)$$

Then the exact bridge gives

$$\boxed{\mathcal{G}_{h_0}(G_X^L) \geq X^{-(\lambda_{\text{sel}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}}) + o(1)} \mathcal{E}_{\text{par}}.} \quad (101)$$

Equation (101) is a conditional composition, not an assertion that any of (97)–(100) holds for the inherited packet. The new singleton gate is exactly λ_{sing} . There is no additional coherence exponent after (100), because the surviving core energy lies in singleton fibers of the complete output. Native-Gram, boundary, and tail comparisons remain separate downstream operations.

5 The sharp rowwise singleton constant

The terminal amplitudes inherited from the literal packet contain one bounded coefficient ζ_m for each source row. This section gives the exact singleton constant that is uniform over that coefficient class. The result is a finite weighted-graph theorem; no distribution of primes is assumed.

Let $\mathcal{M}$ be a finite set of source rows, let $\mathcal{Y}$ be a finite set of output fibers, and let $\mathcal{U}^\circ$ be a finite set of baseline edges with maps

$$m : \mathcal{U}^\circ \rightarrow \mathcal{M}, \quad q : \mathcal{U}^\circ \rightarrow \mathcal{Y}. \quad (102)$$

Assume that a fixed pair (m_0, y) supports at most one edge. Attach to each $u \in \mathcal{U}^\circ$ a nonzero baseline amplitude $c_u \in \mathbb{C}$ and the positive weight

$$w_u := |c_u|^2. \quad (103)$$

The baseline fiber degree is

$$d_y^\circ := |\{u \in \mathcal{U}^\circ : q(u) = y\}|. \quad (104)$$

For a row coefficient vector $\zeta = (\zeta_m)_{m \in \mathcal{M}}$, set

$$x_u(\zeta) := c_u \zeta_{m(u)}. \quad (105)$$

Its physical degree is

$$n_y(\zeta) := |\{u \in \mathcal{U}^\circ : q(u) = y, \zeta_{m(u)} \neq 0\}|. \quad (106)$$

Define

$$D(\zeta) := \sum_{u \in \mathcal{U}^\circ} |x_u(\zeta)|^2, \quad (107)$$

$$D_1^{\text{phys}}(\zeta) := \sum_{\substack{y \in \mathcal{Y} \\ n_y(\zeta)=1}} \sum_{\substack{u \in \mathcal{U}^\circ \\ q(u)=y}} |x_u(\zeta)|^2. \quad (108)$$

The row weights and their structurally singleton portions are

$$W_m := \sum_{\substack{u \in \mathcal{U}^\circ \\ m(u)=m}} w_u, \quad (109)$$

$$W_{m,1} := \sum_{\substack{u \in \mathcal{U}^\circ \\ m(u)=m, d_{q(u)}^\circ=1}} w_u. \quad (110)$$

Assume $\mathcal{U}^\circ \neq \emptyset$, discard the rows with $W_m = 0$, and put

$$\beta_X := \min_{m: W_m > 0} \frac{W_{m,1}}{W_m}. \quad (111)$$

The subscript records the intended arithmetic application; the theorem itself is finite-dimensional.

Theorem 5.1 (Exact rowwise minimax constant). *For every $C > 0$,*

$$\boxed{\inf_{\substack{\zeta \in \mathbb{C}^{\mathcal{M}}, 0 < D(\zeta) \\ |\zeta_m| \leq C \ (m \in \mathcal{M})}} \frac{D_1^{\text{phys}}(\zeta)}{D(\zeta)} = \beta_X.} \quad (112)$$

In particular, every admissible ζ satisfies

$$D_1^{\text{phys}}(\zeta) \geq \beta_X D(\zeta). \quad (113)$$

If $G(\zeta)$ denotes the undeleted grouped energy associated with (105), then

$$G(\zeta) \geq \beta_X D(\zeta). \quad (114)$$

Proof. A baseline singleton edge remains a physical singleton whenever its row coefficient is nonzero. Consequently,

$$D_1^{\text{phys}}(\zeta) \geq \sum_m |\zeta_m|^2 W_{m,1} \geq \beta_X \sum_m |\zeta_m|^2 W_m = \beta_X D(\zeta). \quad (115)$$

This proves the lower bound in (112). The grouped estimate (114) then follows from [theorem 3.1](#).

Choose a row m_0 attaining the minimum in (111). Set $\zeta_{m_0} = C$, and for every other row with positive weight set $\zeta_m = \varepsilon$, where $0 < \varepsilon < C$. Rows of zero weight may be assigned coefficient zero. All baseline edges are then active, so $n_y(\zeta) = d_y^\circ$. Hence the physical singleton energy is exactly the structural singleton energy:

$$\frac{D_1^{\text{phys}}(\zeta)}{D(\zeta)} = \frac{C^2 W_{m_0,1} + \varepsilon^2 \sum_{m \neq m_0} W_{m,1}}{C^2 W_{m_0} + \varepsilon^2 \sum_{m \neq m_0} W_m}. \quad (116)$$

Letting $\varepsilon \rightarrow 0^+$ proves the reverse inequality and therefore the exact identity (112). $\square$

There is an exact rowwise collision-moment form of $W_{m,1}$. Put

$$M^\circ := \max_{y \in \mathcal{Y}} d_y^\circ \quad (117)$$

and, for $r \geq 0$, define

$$J_{m,r} := \sum_{\substack{u \in \mathcal{U}^\circ \\ m(u)=m}} w_u \binom{d_{q(u)}^\circ - 1}{r}. \quad (118)$$

Thus $J_{m,0} = W_m$, and

$$J_{m,1} = \sum_{\substack{u \in \mathcal{U}^\circ \\ m(u)=m}} w_u (d_{q(u)}^\circ - 1) \quad (119)$$

is the baseline rooted collision incidence carried by row m .

Proposition 5.2 (Row collision hierarchy). *For every row m with $W_m > 0$,*

$$\boxed{W_{m,1} = \sum_{r=0}^{M^\circ-1} (-1)^r J_{m,r}.} \quad (120)$$

The corresponding Bonferroni truncations alternate around $W_{m,1}$. In particular,

$$W_{m,1} \geq W_m - J_{m,1}. \quad (121)$$

If for some $0 < \delta \leq 1$,

$$J_{m,1} \leq (1 - \delta)W_m \quad \text{for every } m \text{ with } W_m > 0, \quad (122)$$

then

$$\beta_X \geq \delta \quad \text{and} \quad G(\zeta) \geq \delta D(\zeta) \quad (123)$$

for every admissible coefficient vector with positive energy.

Proof. Apply the binomial identity

$$\mathbf{1}_{\{d=1\}} = \sum_{r=0}^{d-1} (-1)^r \binom{d-1}{r}$$

to $d = d_{q(u)}^\circ$, multiply by w_u , and sum over the edges rooted at row m . This proves (120). Its first odd Bonferroni truncation is (121). Under (122), this gives $W_{m,1} \geq \delta W_m$ on every positive-weight row, hence $\beta_X \geq \delta$. The last assertion follows from [theorem 5.1](#). $\square$

For one distinguished coefficient vector, uniform row control is stronger than necessary. The following weighted criterion records the weaker target.

Corollary 5.3 (Distinguished weighted collision criterion). *For every coefficient vector ζ ,*

$$G(\zeta) \geq \left(D(\zeta) - \sum_m |\zeta_m|^2 J_{m,1} \right)_+. \quad (124)$$

Consequently, if $D(\zeta) > 0$ and

$$\sum_m |\zeta_m|^2 J_{m,1} \leq (1 - \delta)D(\zeta), \quad (125)$$

then

$$G(\zeta) \geq \delta D(\zeta). \quad (126)$$

Proof. Multiply (121) by $|\zeta_m|^2$, sum over m , and use (115) before its second inequality. This gives

$$D_1^{\text{phys}}(\zeta) \geq D(\zeta) - \sum_m |\zeta_m|^2 J_{m,1}.$$

Now apply theorem 3.1 and nonnegativity of $G(\zeta)$. The hypothesis (125) gives the final claim. $\square$

Remark 5.4 (What the minimax theorem does and does not prove). The number β_X is the exact best uniform constant for the *singleton-energy* route over the declared bounded row-coefficient class. If $\beta_X = 0$, the singleton ratio can be arbitrarily small in that class. This does not imply that the grouped energy itself can be made zero: nonsingleton fibers may still contribute constructively. Nor does it give a negative statement about one distinguished coefficient vector. Conversely, a positive value of β_X is a finite-system statement; an asymptotic lower bound for β_X requires separate information about the baseline terminal carrier.

6 Exact factor exchange inside a residual fiber

We work only with the literal fixed-shift carrier inherited from TPC-55–TPC-56. Fix $h_0 \neq 0$, put

$$H := \text{rad } |h_0|,$$

and take X sufficiently large that $L > |h_0|$. A terminal atom is written

$$u = (d\ell, r, j, \gamma), \quad d \in [D, 2D], \quad \ell \in [L, 2L] \cap \mathbb{P},$$

where d, r, dr are squarefree, $(d, r) = 1$, and

$$rp - d\ell j = h_0 \tag{127}$$

for the terminal prime p , with every retained-row, cofactor, residue, profile, and terminal-support condition imposed exactly as in the literal carrier. Its coarse output key is

$$q(u) = (\gamma, s, j), \quad s = dr.$$

No ambient Diophantine solution is called an atom unless it belongs to that exact carrier.

6.1 The factor-swap normal form

Proposition 6.1 (Unique factor exchange). *Let*

$$u_i = (d_i \ell_i, r_i, j, \gamma), \quad i = 1, 2,$$

be two terminal atoms in the same coarse output fiber. There are unique positive, pairwise coprime, squarefree integers a, b, c, e such that

$$\boxed{d_1 = ab, \quad d_2 = ac, \quad r_1 = ce, \quad r_2 = be, \quad s = abce.} \tag{128}$$

They are given by

$$a = (d_1, d_2), \quad b = d_1/a, \quad c = d_2/a, \quad e = s/(abc) = (r_1, r_2). \tag{129}$$

Conversely, any tuple of the form (128) that satisfies both literal terminal equations and all carrier restrictions gives two atoms in one coarse fiber.

Proof. Because s is squarefree and $d_i r_i = s$ with $(d_i, r_i) = 1$, every prime of s lies in exactly one of four disjoint classes: in both d_1, d_2 , in d_1, r_2 , in d_2, r_1 , or in both r_1, r_2 . Their products are respectively a, b, c, e , which proves existence, pairwise coprimality, and (128). The four prime classes are recovered from (d_1, d_2, r_1, r_2) , so the representation is unique. The converse is immediate after the literal support conditions, rather than only the displayed equations, are imposed. $\square$

The normal form may also be viewed relative to one atom. If $d_1 = d$ and $r_1 = r$, a different divisor choice is obtained by choosing $b \mid d$ and $c \mid r$ and exchanging these two coprime factor packets:

$$d_2 = \frac{d}{b}c, \quad r_2 = \frac{r}{c}b. \quad (130)$$

The pair (b, c) is unique once the partner divisor d_2 is known.

Proposition 6.2 (Exchange identity and exact divisibility). *Under the hypotheses of proposition 6.1, the two terminal equations imply*

$$\boxed{e(cp_1 - bp_2) = aj(bl_1 - cl_2)}. \quad (131)$$

Moreover, for $i = 1, 2$,

$$(r_i, j) = (r_i, h_0), \quad (132)$$

and consequently

$$\boxed{\frac{e}{(e, h_0)} \mid bl_1 - cl_2}. \quad (133)$$

Proof. Substitute (128) into the two copies of (127) and subtract:

$$cep_1 - abl_1j = bep_2 - acl_2j.$$

This is (131).

For sufficiently large X , one has $(\ell_i, r_i) = 1$. Indeed, the prime ℓ_i dividing r_i would divide both terms on the left of (127), and hence would divide the fixed nonzero integer h_0 , contrary to $\ell_i \geq L > |h_0|$. Thus $(d_i \ell_i, r_i) = 1$, and reduction of the terminal equation modulo r_i gives

$$j \equiv -h_0(d_i \ell_i)^{-1} \pmod{r_i}.$$

Multiplication by a unit modulo r_i proves (132).

Finally $(a, e) = 1$, so (131) gives $e \mid j(bl_1 - cl_2)$. Since $(e, j) = (e, h_0)$, division by this gcd proves (133). $\square$

6.2 Same-divisor and cross-divisor separation

Corollary 6.3 (Same-divisor collisions require a short cofactor). *Suppose $u_1 \neq u_2$ and $d_1 = d_2 = d$. Then $r_1 = r_2 = r$, $\ell_1 \neq \ell_2$, and*

$$\frac{r}{(r, h_0)} \mid \ell_1 - \ell_2. \quad (134)$$

In particular,

$$\boxed{r \leq HL}. \quad (135)$$

Therefore, on a cofactor block $r \in [R, 2R]$ with $R > HL$, a fixed divisor d contributes at most one terminal atom to any coarse output fiber.

Proof. In the unique factorization of proposition 6.1, equality $d_1 = d_2$ gives $b = c = 1$, $a = d$, and $e = r$. Distinctness then forces $\ell_1 \neq \ell_2$, while (133) gives (134). Since r is squarefree, $(r, h_0) \mid H$, and since two points of $[L, 2L]$ differ by at most L , the asserted bound follows. $\square$

Proposition 6.4 (Large exchanged factors in a balanced block). *Assume in addition that $L > 4D$, that $d_1 \neq d_2$, and that*

$$d_i \in [D, 2D], \quad r_i \in [R, 2R] \quad (i = 1, 2). \quad (136)$$

Then

$$b\ell_1 - c\ell_2 \neq 0 \quad (137)$$

and the exchanged factors in (128) obey

$$\boxed{b^2 \geq \frac{R}{6HL}, \quad c^2 \geq \frac{R}{6HL}.} \quad (138)$$

Equivalently, both exchanged factor packets have size at least

$$\left(\frac{R}{6HL} \right)^{1/2}.$$

Proof. If $b\ell_1 = c\ell_2$, pairwise coprimality of b, c implies $b \mid \ell_2$ and $c \mid \ell_1$. But $b, c \leq 2D < L/2 \leq \ell_i/2$, and the ℓ_i are prime. Hence $b = c = 1$, contrary to $d_1 \neq d_2$. This proves (137).

By (133), squarefreeness of e , and the definition of H ,

$$e \leq H|b\ell_1 - c\ell_2| \leq 2HL(b + c).$$

The ratio $b/c = d_1/d_2$ lies in $[1/2, 2]$. Thus $e \leq 6HLb$ and, symmetrically, $e \leq 6HLc$. Since $r_2 = be \geq R$ and $r_1 = ce \geq R$, one also has $e \geq R/b$ and $e \geq R/c$. Combining the two pairs of inequalities gives (138). $\square$

6.3 The exact partner candidate set

Fix a terminal atom $u = (d\ell, r, j, \gamma)$. Define $\mathfrak{C}(u)$ to be the set of triples (b, c, ℓ') for which

(i) $b \mid d$, $c \mid r$, and $(b, c) \neq (1, 1)$;

(ii)

$$d' = \frac{d}{b}c, \quad r' = \frac{r}{c}b$$

are integers in the declared divisor and cofactor supports;

(iii) ℓ' is in the exact declared source-prime support and the atom

$$u' = (d'\ell', r', j, \gamma)$$

belongs to the literal terminal-admissible universe, including all retained-pair, residue, cutoff, and carrier restrictions.

The terminal prime of u' is not a free coordinate: it is exactly

$$p' = \frac{d'\ell'j + h_0}{r'}. \quad (139)$$

Proposition 6.5 (Exact partner parameterization). *The map*

$$u' \longmapsto \left(\frac{d}{(d, d')}, \frac{d'}{(d, d')}, \ell' \right) \quad (140)$$

is a bijection from the distinct-divisor terminal partners of u in its full coarse output fiber onto $\mathfrak{C}(u)$. Same-divisor partners are exactly the further source primes on the single determinant ladder with $d' = d$ and $r' = r$.

Proof. For a partner u' , apply [proposition 6.1](#) with $d_1 = d, r_1 = r$. It gives the unique divisors $b = d/(d, d')$ and $c = d'/(d, d')$, and (130); literal membership gives all the conditions defining $\mathfrak{C}(u)$. Conversely, every element of $\mathfrak{C}(u)$ defines a literal terminal atom with $d'r' = dr = s$, the same j, γ , and hence the same output key. Uniqueness of the factor swap and of the source row proves bijectivity. The case $d' = d$ has $b = c = 1$ and is precisely the fixed-divisor ladder. $\square$

Assume now that the root cofactor satisfies $r \in [R, 2R]$, and constrain its partners to the same block $r' \in [R, 2R]$. Define

$$\mathfrak{C}_R(d, r) := \left\{ (b, c) : \exists \ell' \begin{array}{l} (b, c, \ell') \in \mathfrak{C}(u), \\ r' = (r/c)b \in [R, 2R] \end{array} \right\}. \quad (141)$$

The fixed-divisor ladder spacing gives the deterministic degree estimate

$$\#\{u' \neq u : q(u') = q(u), r' \in [R, 2R]\} \leq \left\lfloor \frac{HL}{R} \right\rfloor + \left(1 + \left\lfloor \frac{HL}{R} \right\rfloor\right) |\mathfrak{C}_R(d, r)|. \quad (142)$$

When $d' \neq d$, every pair counted by $\mathfrak{C}_R(d, r)$ additionally satisfies (138). This is sharper structural information than the untruncated divisor bound $|\mathfrak{C}_R(d, r)| \leq \tau(d)\tau(r)$, although it does not by itself force the candidate set to be empty.

Corollary 6.6 (High-cofactor structural exclusion). *Under the hypotheses above, if*

$$R > 24HLD^2, \quad (143)$$

then every coarse terminal fiber restricted to $d \in [D, 2D]$ and $r \in [R, 2R]$ contains at most one atom.

Proof. The hypothesis implies $R > HL$, so [corollary 6.3](#) excludes a same-divisor pair. For a cross-divisor pair, $b \leq d_1 \leq 2D$, whereas (138) and (143) give $b^2 > 4D^2$, a contradiction. $\square$

There is also a full-fiber version, with only a harmless change of the constant. It is useful when a core cofactor block is used merely to find energy and no block-restricted output is introduced.

Corollary 6.7 (A high-cofactor atom is a full-fiber singleton). *Assume $L > 4D$. Let u_1 be a terminal atom with $d_1 \in [D, 2D]$ and $r_1 \in [R, 2R]$, while every possible source divisor in its full literal carrier remains in $[D, 2D]$. If*

$$R > 48HLD^2, \quad (144)$$

then u_1 has no distinct terminal partner anywhere in its full coarse output fiber.

Proof. Every partner has $r_2/r_1 = d_1/d_2 \in [1/2, 2]$, hence $r_2 \in [R/2, 4R]$. A same-divisor partner is excluded by [corollary 6.3](#). For a cross-divisor partner, the proof of [proposition 6.4](#) still gives $e \leq 6HLb$, while $r_2 = be \geq R/2$. Therefore

$$b^2 \geq \frac{R}{12HL} > 4D^2,$$

contradicting $b \leq d_1 \leq 2D$. $\square$

The last corollary is a support theorem only. In fact, the inherited large-prime carrier has a global cofactor ceiling that is disjoint from this high-cofactor regime; see [equations \(168\)](#) and [\(169\)](#). Thus the corollary is a sharp structural boundary, not an activation mechanism for the present physical packet.

Finally, the factor geometry localizes possible partners up to a constant scale enlargement. If one atom has $r_1 \in [R, 2R]$, then every atom in its full coarse fiber satisfies

$$\frac{1}{2} \leq \frac{r_2}{r_1} = \frac{d_1}{d_2} \leq 2, \quad r_2 \in [R/2, 4R]. \quad (145)$$

Thus a singleton audit for a core block needs to inspect only its constant-factor saturation, not remote cofactor scales. This observation does not license a lower bound for a block-restricted output to be returned to the full sum; it merely identifies the complete set in which a full fiber partner can occur.

7 Orbit congruence and the two-affine-prime reduction

We retain the literal hypotheses and notation of [section 6](#). In particular $h_0 \neq 0$ is fixed, $H = \text{rad } |h_0|$, X is sufficiently large that $L > |h_0|$, and no solution is counted unless it belongs to the exact retained carrier.

7.1 Exact orbit congruence

Lemma 7.1 (The terminal gcd is fixed by the intercept). *For every terminal atom $u = (d\ell, r, j, \gamma)$,*

$$(d\ell, r) = 1, \quad j \equiv -h_0(d\ell)^{-1} \pmod{r}, \quad \boxed{(r, j) = (r, h_0)}. \quad (146)$$

Since r is squarefree, the common gcd in (146) divides H .

Proof. The literal carrier gives $(d, r) = 1$. If the source prime ℓ divided r , then both rp and $d\ell j$ would be divisible by ℓ , so the terminal equation would imply $\ell \mid h_0$, impossible for $\ell \geq L > |h_0|$. Hence $(d\ell, r) = 1$. Reducing $rp - d\ell j = h_0$ modulo r gives the displayed congruence. Multiplication by a unit modulo r preserves the gcd with r , so $(r, j) = (r, h_0)$. The final assertion follows from squarefreeness. $\square$

Corollary 7.2 (Orbit-time sparsity for a fixed row and cofactor). *Fix d, r, ℓ satisfying the literal source and cofactor conditions. In an orbit interval of diameter at most $C_J J$, the terminal equality can hold for at most*

$$1 + \left\lfloor \frac{C_J J}{r} \right\rfloor \quad (147)$$

integer values of j . Intersecting with the declared orbit residue class, profiles, or terminal-prime condition can only decrease this number.

Proof. By [lemma 7.1](#), every possible j lies in one fixed residue class modulo r . The elementary spacing bound in an interval of diameter $C_J J$ proves the result. $\square$

The congruence is stronger than the necessary statement $(r, j) \mid h_0$: on the literal terminal support the gcd is exactly (r, h_0) . It does not assert that the unique residue class contains a retained j , or that the resulting quotient is prime.

7.2 A simultaneous two-row orbit class

Use the factor-exchange notation

$$d_1 = ab, \quad d_2 = ac, \quad r_1 = ce, \quad r_2 = be,$$

with a, b, c, e pairwise coprime. For fixed $(a, b, c, e, \ell_1, \ell_2)$, the two terminal equalities require

$$\begin{aligned} j &\equiv -h_0(ab\ell_1)^{-1} \pmod{ce}, \\ j &\equiv -h_0(ac\ell_2)^{-1} \pmod{be}. \end{aligned} \quad (148)$$

All inverses exist by [lemma 7.1](#).

Proposition 7.3 (Exact CRT compatibility and joint orbit spacing). *The two congruences in (148) are compatible if and only if*

$$\boxed{\frac{e}{(e, h_0)} \mid b\ell_1 - c\ell_2.} \quad (149)$$

When compatible, they define one residue class modulo

$$\text{lcm}(ce, be) = bce. \quad (150)$$

Consequently an orbit interval of diameter $C_J J$ contains at most

$$1 + \left\lfloor \frac{C_J J}{bce} \right\rfloor \quad (151)$$

simultaneously admissible integer times. A fixed inherited orbit class can only thin this set further.

Proof. The two residue classes must agree modulo $(ce, be) = e$. Multiplying their equality modulo e by the available units shows that compatibility is equivalent to

$$e \mid h_0(b\ell_1 - c\ell_2),$$

which is exactly (149). The Chinese remainder theorem then gives one class modulo the least common multiple bce , and the spacing estimate follows. $\square$

This is an orbit-counting statement, not a prime-counting theorem. Once j is chosen, both terminal candidates are determined, but the CRT does not prove either candidate prime.

7.3 The determinant ladder

Fix a coarse output $y = (\gamma, s, j)$ and an admissible divisor $d \mid s$, and put $r = s/d$. The source and terminal primes must solve

$$rp - dj\ell = h_0. \quad (152)$$

Let

$$g := (r, dj) = (r, j), \quad (153)$$

where the second equality uses $(d, r) = 1$.

Proposition 7.4 (Complete nonproportional affine reduction). *Equation (152) has an integral solution if and only if $g \mid h_0$. If (ℓ_0, p_0) is one solution, every integral solution is uniquely*

$$\boxed{\ell(t) = \ell_0 + \frac{r}{g}t, \quad p(t) = p_0 + \frac{dj}{g}t, \quad t \in \mathbb{Z}.} \quad (154)$$

The two affine forms have nonzero cross determinant

$$\frac{dj}{g}\ell_0 - \frac{r}{g}p_0 = -\frac{h_0}{g} \neq 0. \quad (155)$$

On the literal terminal support,

$$g = (r, h_0) \mid H. \quad (156)$$

Proof. The gcd criterion for the linear Diophantine equation is $g \mid h_0$. Differences of two solutions satisfy $r\Delta p = dj\Delta\ell$. After division by g , the two remaining coefficients are coprime, so $\Delta\ell = (r/g)t$ and $\Delta p = (dj/g)t$, proving (154). Substitution of the base solution gives (155). Finally, lemma 7.1 gives $g = (r, j) = (r, h_0)$, and squarefreeness of r gives the divisibility by H . $\square$

Let $\mathfrak{T}_{\gamma,s,j,d} \subset \mathbb{Z}$ be the exact set of parameters t in (154) for which

- (i) $\ell(t)$ belongs to the declared source-prime row support;
- (ii) the pair $(d\ell(t), r)$, the orbit time j , and the opposite row γ satisfy every retained literal carrier condition;
- (iii) $p(t)$ is in the declared terminal-prime support and its shaped terminal amplitude is nonzero.

For terminal-admissible occupancy one does not additionally delete a zero source coefficient; for physical occupancy one intersects this set with the exact nonzero coefficient support. With that convention, the fixed divisor occupancy is the identity

$$N_{\gamma,s,j,d} = \#\mathfrak{T}_{\gamma,s,j,d} = \sum_{t \in \mathbb{Z}} \mathbf{1}_{\mathfrak{T}_{\gamma,s,j,d}}(t). \quad (157)$$

This definition retains the literal carrier and does not replace it by all prime values of the two affine forms.

Corollary 7.5 (Deterministic ladder capacity). *For source-prime support inside $[L, 2L]$,*

$$N_{\gamma,s,j,d} \leq 1 + \left\lfloor \frac{Lg}{r} \right\rfloor \leq 1 + \left\lfloor \frac{LH}{r} \right\rfloor. \quad (158)$$

In particular, $r > LH$ forces $N_{\gamma,s,j,d} \leq 1$.

Proof. The first affine form in (154) has spacing r/g , and $[L, 2L]$ has diameter L . The second inequality uses (156). Every further literal condition only deletes parameters. $\square$

7.4 What prime input remains

After the deterministic reductions, actual occupancy on one fixed divisor is a simultaneous-primality problem for the two nonproportional affine forms

$$\ell_0 + \frac{r}{g}t, \quad p_0 + \frac{dj}{g}t, \quad (159)$$

whose determinant is the fixed nonzero integer $-h_0/g$. The determinant rules out proportional forms and exposes the local congruence data. It does not provide a lower bound for prime values. In particular:

- (i) a lower bound on an individual growing ladder is a two-linear-form prime problem and is not supplied by the elementary geometry;
- (ii) an upper-bound sieve may thin the ambient ladder by logarithmic factors, but it neither proves terminal activation nor compares a collision second moment to the physical terminal energy;
- (iii) averaging over d, r, j, s may be more tractable than one fixed ladder, but uniformity in the moving coefficients, literal masks, residue classes, and physical coefficient weights requires a separate analytic theorem.

Thus [proposition 7.4](#) is an exact arithmetic reduction of the next gate. It is not a singleton-density theorem, a parity-breaking estimate, or a fixed- h_0 prime-pair lower bound.

8 Occupancy barrier, endpoint ledger, and the next gate

The preceding sections give exact lower bounds once a positive fraction of the terminal diagonal lies in singleton fibers of the complete physical output. They do not supply that fraction. This section records why the existing support bounds do not imply it, states the precise weighted factor-exchange estimate still missing, and places that estimate in the inherited $1/400$ ledger.

8.1 Why a multiplicity upper bound is not singleton occupancy

The TPC-55 determinant-ladder argument gives an upper bound for the number of terminal atoms in one residual fiber. Such a bound is useful for a post-terminal transversal cover, but it points in the wrong direction for the undeleted singleton route.

Proposition 8.1 (Sharp occupancy barrier). *For every integer $M \geq 2$ and every $E > 0$, there is a finite fiber system and a vector x such that*

$$D(x) = E, \quad \max_y n_y(x) \leq M, \quad D_1(x) = G(x) = 0. \quad (160)$$

Consequently no positive lower bound for D_1/D , or for G/D , follows from a finite maximal-fiber bound alone. The conclusion remains true when the maximal degree is 2.

Proof. Use one fiber with two coordinates

$$x_{u_1} = \sqrt{E/2}, \quad x_{u_2} = -\sqrt{E/2}.$$

Its active degree is 2, its singleton energy is zero, and its grouped sum vanishes. This is also the case $D_0 = I_0 = E$ of [proposition 3.3](#). $\square$

The obstruction is not cancellation-specific. Even if both coordinates in every doubleton have the same phase, the singleton energy is still zero. Thus replacing the grouped square by a random-phase heuristic would not prove the required occupancy statement. Likewise, terminal activation $\mathcal{D}_R^{\text{core}} > 0$ says only that some terminal energy survives; it does not say whether that energy lies in degree-one or multiple fibers.

The rowwise minimax theorem makes the same boundary coefficient-sensitive. For a declared baseline carrier, the best constant uniform over the bounded row-coefficient class is

$$\beta_X = \min_{m: W_m > 0} \frac{W_{m,1}}{W_m}.$$

If one positive-weight row has no structurally singleton edge, then $\beta_X = 0$. An upper bound on every fiber degree does not exclude this case. Conversely, $\beta_X = 0$ is only a failure of the uniform singleton route; it is not a proof that the distinguished grouped vector vanishes.

8.2 The current endpoint scales

The literal scale dictionary inherited from TPC-53 and retained in TPC-55-56 is

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad D = X^{10049/52500+o(1)}, \quad (161)$$

and therefore

$$L = Q/D = X^{99979/210000+o(1)}. \quad (162)$$

For a dyadic cofactor block $r \asymp R = X^{\theta_R+o(1)}$, the inherited coefficient-blind capacity bound is

$$M_X(R) \ll X^{o(1)} \left(1 + \frac{L}{R} \right), \quad \mu_{\text{cap}} = \left(\frac{99979}{210000} - \theta_R \right)_+. \quad (163)$$

The enlarged partner range $[R/2, 4R]$ is the union of three dyadic blocks with scales comparable to R . Hence

$$\max_y d_y^+(R) \ll X^{o(1)} \left(1 + \frac{L}{R}\right) \quad (164)$$

under the same polynomial-size residual-label hypothesis as TPC-55. The constant-factor enlargement changes no exponent.

At the older cofactor scale $R \asymp J$, one has

$$\frac{L}{R} = X^{15077/105000+o(1)}, \quad \mu_{\text{cap}} = \frac{15077}{105000}. \quad (165)$$

This support-only exponent exceeds the nominal endpoint allowance by

$$\frac{15077}{105000} - \frac{1}{400} = \frac{29629}{210000} > 0. \quad (166)$$

The factor-exchange geometry also gives an exact support theorem in a much higher cofactor regime: by [corollary 6.7](#), every core atom is a singleton in its complete terminal fiber if

$$R > 48HLD^2, \quad H = \text{rad } |h_0|. \quad (167)$$

Here $L > 4D$ holds for all sufficiently large X at the declared scales, and $H = X^{o(1)}$ because h_0 is fixed. Let N_+ denote the inherited maximum positive target size. The large-prime descent imposes the stronger global range restriction [\[2\]](#)

$$p > \sqrt{N_+}, \quad r = \frac{mj + h_0}{p} < \sqrt{N_+} = X^{1/2+o(1)}. \quad (168)$$

On the other hand,

$$LD^2 = X^{180371/210000+o(1)}, \quad \frac{\sqrt{N_+}}{HLD^2} = X^{-75371/210000+o(1)}. \quad (169)$$

Consequently no terminal atom in the inherited physical carrier can lie in a block satisfying [\(167\)](#) once X is large. The high-cofactor theorem is a valid structural boundary, but its physical activation door is closed for the present packet.

For comparison, at the older scale $R \asymp J$,

$$\frac{R}{HLD^2} = X^{-55273/105000+o(1)}. \quad (170)$$

Thus even before using the global ceiling, the old block misses the threshold by a fixed positive power.

For $R \gg L$, the capacity is only $X^{o(1)}$; nevertheless a carrier made entirely of doubletons still has zero singleton energy. Thus moving to a balanced block removes the positive-power multiplicity cost of a transversal cover but does not prove the occupancy estimate needed here. Primality may thin the ambient ladder by logarithmic factors; the existing upper-bound sieve does not turn this support count into a weighted degree-one lower bound.

8.3 The exact missing weighted factor-exchange estimate

The weakest first-moment statement needed for the distinguished vector is not an unweighted count. By [\(92\)](#), it is precisely

$$\boxed{\sum_m |\zeta_m|^2 \mathfrak{J}_{m,1}^{R,+}(\zeta) \leq (1 - \delta_R) \sum_m |\zeta_m|^2 W_{m,R}, \quad \delta_R > 0.} \quad (171)$$

Every pair counted on the left satisfies the full literal system

$$\begin{aligned} u = (d\ell, r, j, \gamma), \quad v = (d'\ell', r', j, \gamma), \quad dr = d'r', \quad u \neq v, \\ pr - d\ell j = h_0, \quad p'r' - d'\ell'j = h_0, \\ R \leq r \leq 2R, \quad R/2 \leq r' \leq 4R, \end{aligned} \quad (172)$$

with both terminal primes, all squarefree/coprime conditions, the literal mask, every profile cutoff, and the actual row coefficients retained. The energy weight is attached to the root u :

$$|x_u|^2 = |c_u|^2 |\zeta_{m_u}|^2. \quad (173)$$

Neither the partner count nor the root count may replace this weight without a separate comparison theorem.

The coefficient-independent baseline majorant gives the stronger sufficient target

$$\boxed{\sum_m |\zeta_m|^2 J_{m,1}^{R,+} \leq (1 - \delta_R) \sum_m |\zeta_m|^2 W_{m,R}.} \quad (174)$$

A still stronger, coefficient-uniform rowwise target is

$$J_{m,1}^{R,+} \leq (1 - \delta_R) W_{m,R} \quad (m : W_{m,R} > 0). \quad (175)$$

The implications are

$$(175) \implies (174) \implies (171) \implies \mathcal{G}_{h_0}(G_X^L) \geq \delta_R \mathcal{D}_R^{\text{core}}. \quad (176)$$

On the admissible inherited carrier, no estimate in this chain is currently proved asymptotically for the $R \asymp J$ packet. The first target is suitable for a theorem uniform over the bounded coefficient class; the middle target may exploit the distinguished coefficient energy; the last is the exact weakest target for the first rooted-moment bridge.

If

$$\delta_R = X^{-\lambda_{\text{sing}} + o(1)}, \quad (177)$$

then λ_{sing} is the singleton-occupancy loss. Together with the exact activation chain in (101), a direct already-reassembled fixed- h_0 route must eventually satisfy

$$\boxed{\lambda_{\text{sel}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} + \lambda_{\text{Gram}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}} \leq \frac{1}{400}.} \quad (178)$$

No separate coherence loss is inserted after the singleton estimate. No frozen-shift localization or projective-reassembly loss is inserted in the direct physical route either. If a different proof passes through those interfaces, their losses must be restored explicitly. At equality in (178), an error already of order $X^{-1/400+o(1)}$ need not be absorbable; strict slack or a closed constant-level comparison is the safe endpoint condition.

8.4 Six-point physical audit

The proposed singleton bridge passes the following audit.

- (A1) *Literal coefficients.* The terminal amplitude is the exact $x_u = (D_{h_0} G_X^L)_u$. Its row-factorized form $x_u = c_u \zeta_{m_u}$ retains the common output factor, source shapes, binary mask, profiles, cofactor sign, and shaped terminal logarithm. The collision statistic is weighted by $|x_u|^2$, not by a proxy counting measure.
- (A2) *Fixed physical shift.* Both equations in (172) use the one specified h_0 . No averaged or frozen intercept is introduced.

- (A3) *Physical atomic normalization.* The pre-terminal, core terminal, collision, and grouped energies all use the inherited raw counting-measure atoms. Repetitions in the cofactor label remain distinct, and the terminal multiplier is not hidden in a modified input norm.
- (A4) *Canonical representation.* Collisions are defined by the complete output key $q(u) = (\gamma, d_m r, j)$, with the terminal-prime coordinate resolved exactly as in TPC–55. The core and enlarged block are deterministic cofactor ranges. No favorable rank, phase sector, projective layer, or ambient prime ladder is selected.
- (A5) *Actual active support.* The baseline terminal carrier and the physically active carrier are kept distinct. The exact statistic $\mathfrak{J}_{m,1}^{R,+}(\zeta)$ uses active degrees, while $J_{m,1}^{R,+}$ is explicitly labeled as a support-only majorant. The enlarged-block lemma accounts for every full-output partner of a core root.
- (A6) *Endpoint budget.* Block occupancy, terminal activation, and singleton occupancy have separate exponents. Native-Gram, boundary, and tail losses remain visible in (178). No support-only multiplicity estimate is entered as a positive singleton fraction, and the high-cofactor support theorem is not used outside its literal range.

8.5 Claim table

Statement	Status
Singleton floor and rooted collision expansion	Unconditional finite-dimensional theorems (L0).
Core-to-full collision bridge	Unconditional finite-dimensional lower bound for the undeleted grouped vector (L0).
Sharp rowwise minimax constant	Exact finite weighted-graph theorem for the bounded row-coefficient class (L0).
Prime-resolved literal row/fiber representation	Exact on the inherited fixed- h_0 terminal support under the TPC–55 injection and TPC–56 gauge factorization (L1).
Enlarged-block partner localization	Exact factor-exchange consequence of $dr = d'r'$ and $d, d' \in [D, 2D]$; no arithmetic occupancy assertion (L1).
High-cofactor full-fiber singleton exclusion	Exact support theorem under $R > 48HLD^2$ (L1); the inherited $r < X^{1/2+o(1)}$ ceiling makes this range physically empty.
Weighted physical incidence identity	Exact specialization to the distinguished physical coefficient vector (L1).
Positive singleton fraction on a cofactor core	Conditional on (171); not proved.
Core block occupancy and terminal activation	Separate hypotheses; not proved here.
Undeleted fixed- h_0 residual lower square	Follows conditionally from the preceding three gates; no unconditional asymptotic lower bound is claimed.
Native affine Gram, boundary, and tail reconnection	Not proved.
Parity improvement or twin-prime consequence	Not claimed.

The level terminology is therefore conservative. L0 denotes exact finite-dimensional identities and extremal laws. L1 denotes exact specialization to the declared literal support and factor-exchange geometry. An L2 advance would require a quantitative estimate such as (171), together with activation and every downstream comparison needed to reconnect to the parent physical correlation. No result in the current paper is labeled L2.

8.6 The next gate

The factor-exchange localization leaves one closed structural shortcut and four live analytic or negative routes.

- Door A.** *Closed high-cofactor shortcut.* The support theorem would give $\delta_R = 1$ above $48HLD^2$, but equations (168) and (169) show that the inherited carrier contains no such terminal atom. This door is closed for the current packet. Reopening it would require a different large-prime decomposition or physical carrier, not merely a new dyadic choice of R .
- Door B.** *Distinguished weighted incidence.* Estimate the exact left side of (171), retaining the root energy $|c_u \zeta_{m_u}|^2$. This is the weakest first-moment target and may exploit concentration of the actual coefficients away from collision-rich rows.
- Door C.** *Uniform row factor exchange.* Prove (175) by analyzing the coupled system (172) for each source row. This would give a theorem uniform over the inherited bounded coefficient class, but is strictly stronger than Door B.
- Door D.** *Higher rooted moments.* If the first-moment truncation is too crude, define core-rooted moments with $\binom{n^{\text{phys}}_{q(u)} - 1}{r}$. The exact Bonferroni hierarchy of theorem 3.2 can then certify singleton energy through an odd higher truncation. Each new moment requires its own weighted factor-exchange estimate.
- Door E.** *Saturation or route closure.* If none of the preceding estimates can hold, construct literal carriers for which the weighted singleton constant tends to zero, or prove that every available support-only estimate is saturated by collision-rich rows. Such a theorem would close the singleton door without asserting cancellation of the distinguished full output.

Door A records a closed structural boundary. Door B is the economical live analytic target. The enlarged block shows that Door B is local in cofactor scale, but the occupancy barrier shows that it is not a consequence of the known ladder capacity bound. A positive estimate, together with core terminal activation, would give the first undeleted fixed- h_0 coherent lower square in this route; a sharp negative estimate would identify precisely why singleton occupancy cannot carry the endpoint argument.

Conclusion. The undeleted terminal square has an exact phase-blind floor: every unit of core energy whose full physical fiber is a singleton survives unchanged. All partners of a cofactor-core atom lie in one constant-factor enlarged block, and the resulting obstruction is the explicit coefficient-weighted factor-exchange incidence (171). The high-cofactor regime is disjoint from the inherited carrier; on its admissible cofactor range, the current support theory bounds how many partners may occur but does not show that a positive weighted fraction has none. Terminal activation, weighted singleton occupancy, native-Gram comparison, boundary control, and tail reconnection therefore remain separate gates, all strictly before any parity-sensitive or twin-prime conclusion.

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